Physical Distancing to Prevent Influenza-Like-Illness on College Campuses: the eX-FLU Trial

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UNC Biostatistics

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eX-FLU Trial

- **eX-FLU**: trial to evaluate a physical distancing intervention on a college campus during flu season (Aiello et al., 2016; Zivich et al., 2020)

Design and methods of a social network isolation study for reducing respiratory infection transmission: The eX-FLU cluster randomized trial

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- **Central question**: does the encouragement-to-isolate intervention prevent ILI infection?

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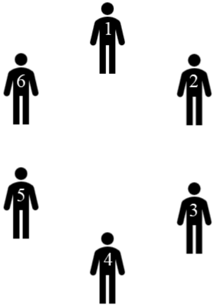
Transmission of Influenza-Like-Illness



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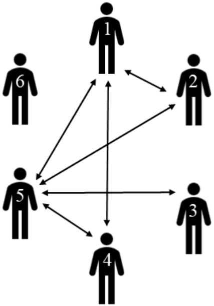

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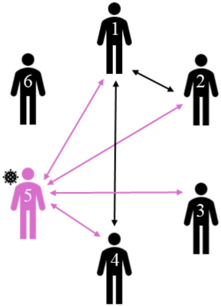

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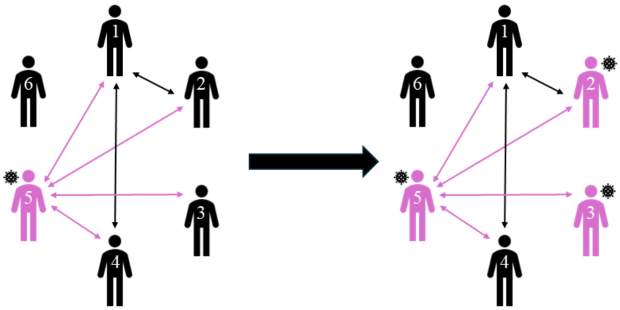
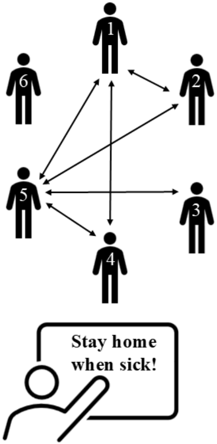


Example I: Intervention Affects Network and Infections



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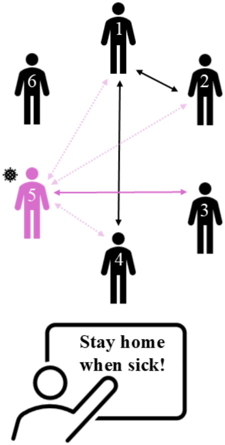

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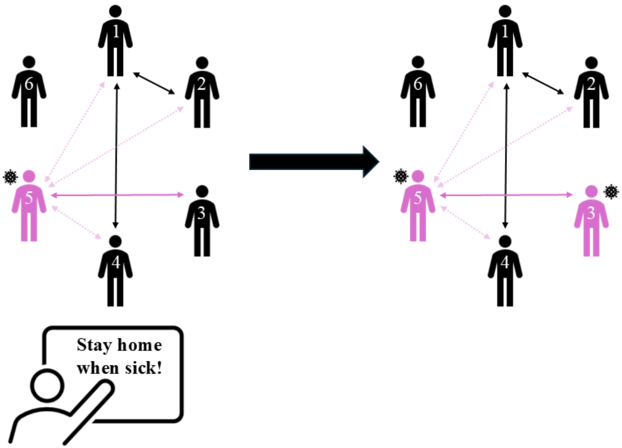
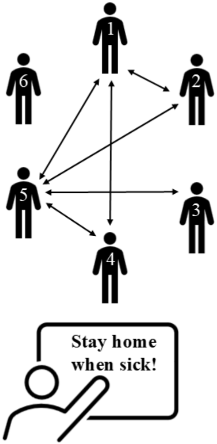


Example II: Intervention Affects Network, Not Infections



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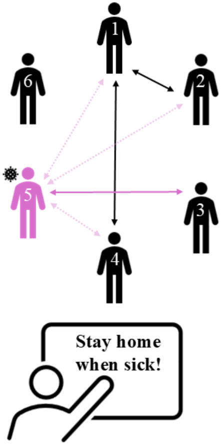

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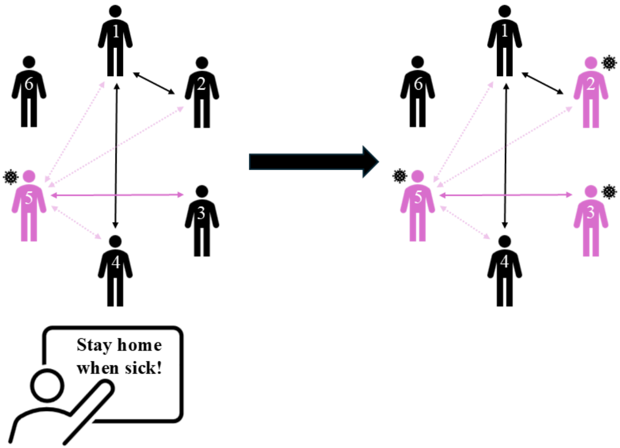

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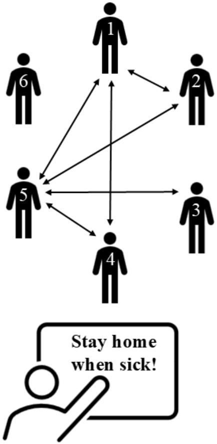

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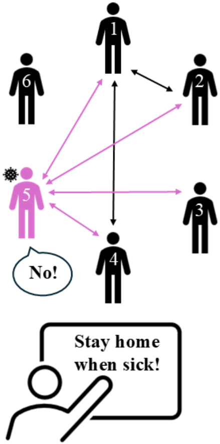

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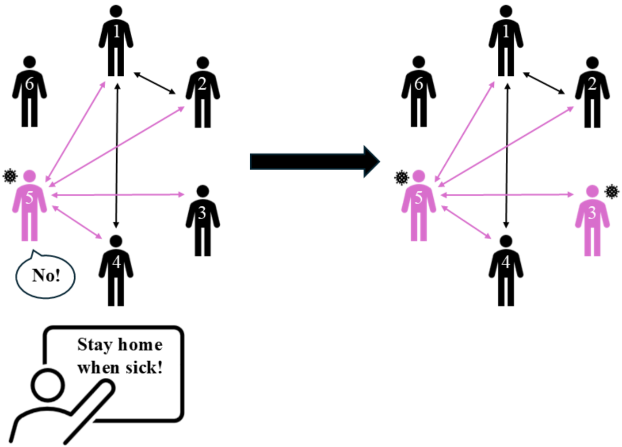
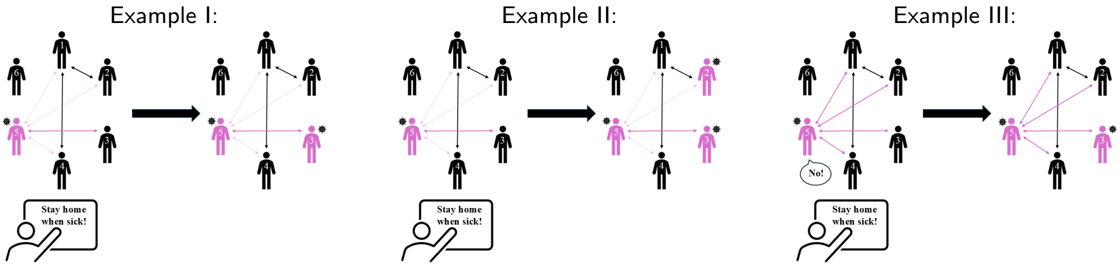


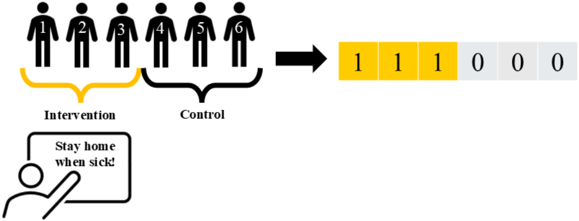
Central Question: Does the Intervention Prevent ILI Infection?



In Examples I and III, the answer is yes

eX-FLU Observed Data

- Baseline randomization assignments $\mathbf{Z} = (Z_1, \ldots, Z_n) \in \mathcal{Z}$, for $Z_i = \mathbb{1}(\text{student } i \text{ gets intervention})$



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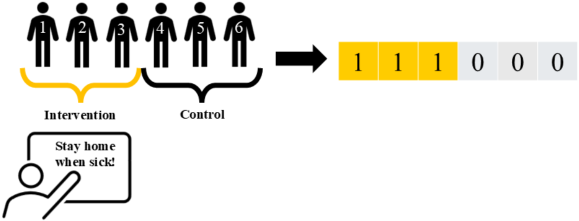

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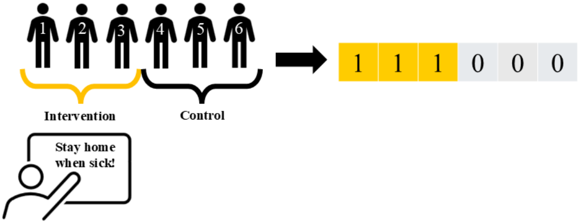

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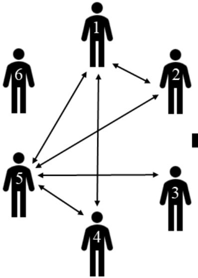

	1	2	3	4	5	6
1	0	1	0	1	1	0
2	1	0	0	0	1	0
3	0	0	0	0	1	0
4	1	0	0	0	1	0
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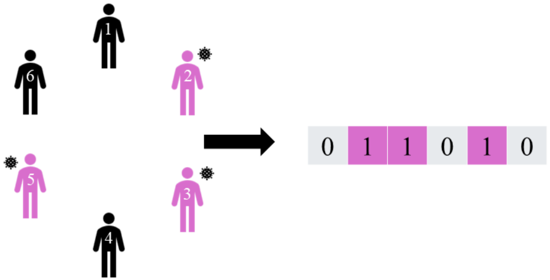

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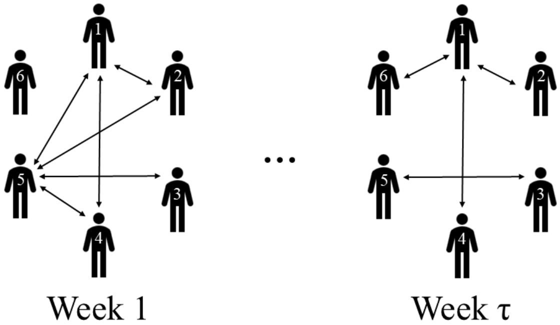
- network history: $\overline{\mathbf{A}} = \{\mathbf{A}^k\}_{k=1}^{\tau}$



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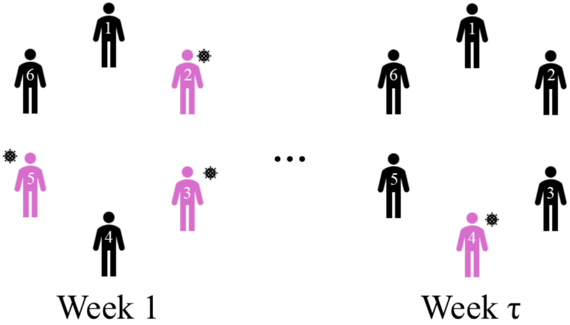

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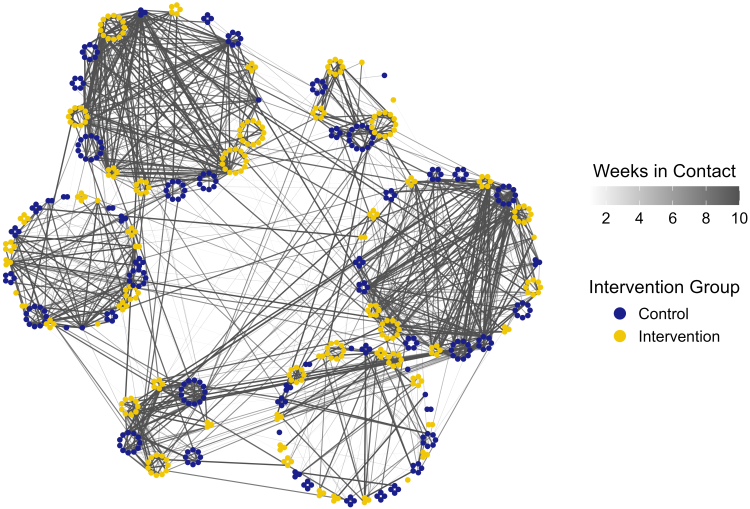

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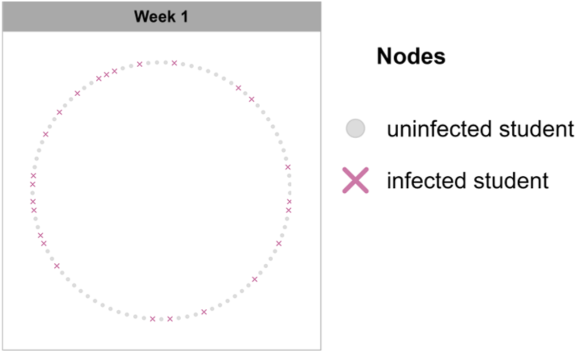

(93 out of 579 students with at least one infection)

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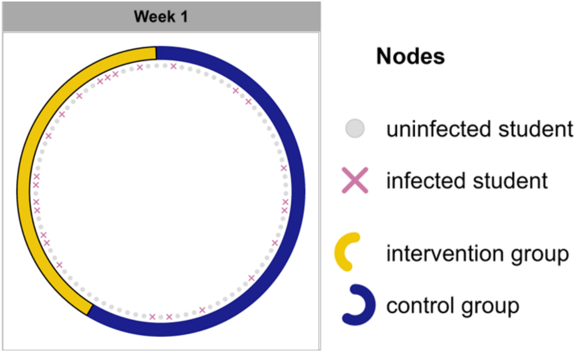

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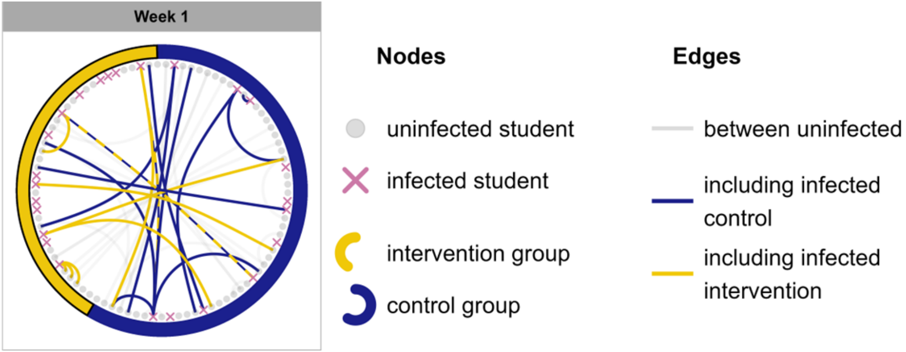

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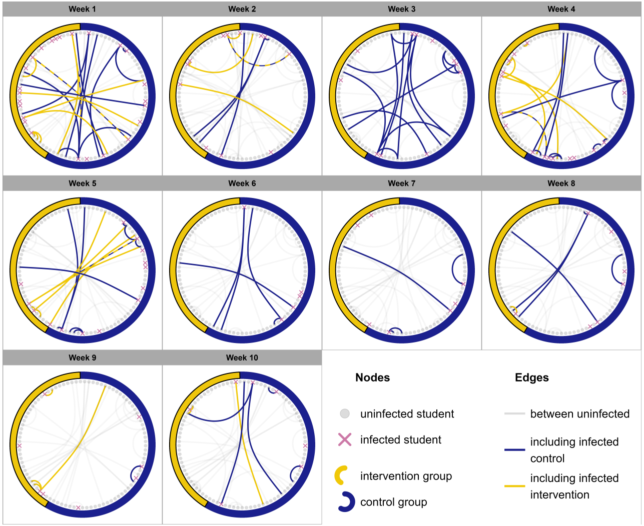

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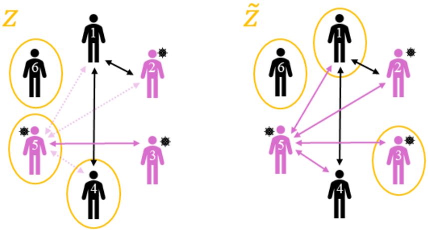
- Assume **causal consistency**:

$$\overline{\mathbf{A}} = \overline{\mathbf{A}}(\mathbf{Z}) \text{ and } \overline{\mathbf{Y}} = \overline{\mathbf{Y}}(\mathbf{Z})$$

eX-FLU Null Hypotheses

$$H_0^Y : \overline{\mathbf{Y}}(\mathbf{z}) = \overline{\mathbf{Y}}(\widetilde{\mathbf{z}}) \text{ for all } \mathbf{z}, \widetilde{\mathbf{z}} \in \mathcal{Z}$$

("the intervention has no effect on the infections")



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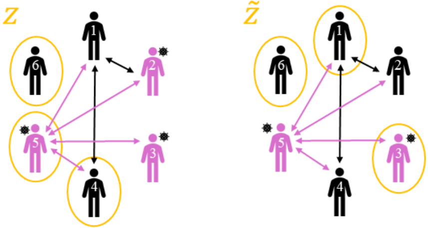

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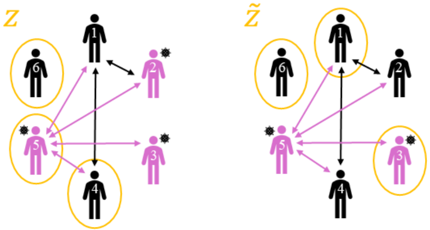

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Previous analyses of the eX-FLU trial (Alexandria et al., 2023) tested $H_0^\sharp$, but no analyses have tested H_0^Y

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Randomization	Networks	Infections	Test Statistic
$\mathbf{z}_1$			
$\mathbf{z}_2$			
$\ldots$			
$\mathbf{z}_{\lvert\mathcal{Z}\rvert}$			

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$\mathbf{z}_2$	$\overline{\mathbf{A}}(\mathbf{z}_2) =?$	$\overline{\mathbf{Y}}(\mathbf{z}_1) =?$	$T(\mathbf{z}_1, ?, ?)$
$\dots$	$\dots$	$\dots$	$\dots$
$\mathbf{z}_{\lvert\mathcal{Z}\rvert}$	$\overline{\mathbf{A}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) =?$	$\overline{\mathbf{Y}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) =?$	$T(\mathbf{z}_{\lvert\mathcal{Z}\rvert}, ?, ?)$

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  - This is relatively easy for a **sharp** null like $H_0^\sharp$
  - This is harder to do for a **nonsharp** null like H_0^Y

Randomization	Networks	Infections	Test Statistic
$\mathbf{z}_1$	$\overline{\mathbf{A}}(\mathbf{z}_1) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_1) = ?$	$T(\mathbf{z}_1, ?, ?)$
$\mathbf{z}_2$	$\overline{\mathbf{A}}(\mathbf{z}_2) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_1) = ?$	$T(\mathbf{z}_1, ?, ?)$
$\ldots$	$\ldots$	$\ldots$	$\ldots$
$\mathbf{z}_{\lvert\mathcal{Z}\rvert}$	$\overline{\mathbf{A}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) = ?$	$T(\mathbf{z}_{\lvert\mathcal{Z}\rvert}, ?, ?)$

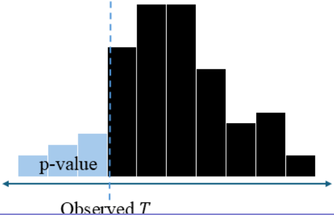
Testing a Null Hypothesis H_0

"How unlikely are the observed data under H_0?" We can answer this using a **randomization test** (Fisher, 1936; Alexandria et al., 2023; Zhang and Zhao, 2023; Ritzwoller et al., 2024)

- Choose a test statistic $T = T(\mathbf{Z}, \overline{\mathbf{A}}, \overline{\mathbf{Y}})$
- Find the distribution of T under H_0
  - This is relatively easy for a **sharp** null like $H_0^\sharp$
  - This is harder to do for a **nonsharp** null like H_0^Y
- Compute a p-value: $\rho^\sharp = F_T(T|H_0^\sharp)$, where

$$F_T(t|H_0^\sharp) = \Pr\left(T \leq t|H_0^\sharp\right)$$

is the **cumulative distribution function** (CDF) of T



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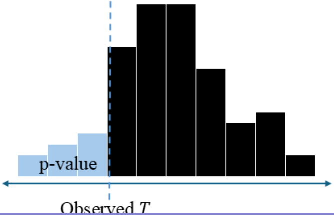
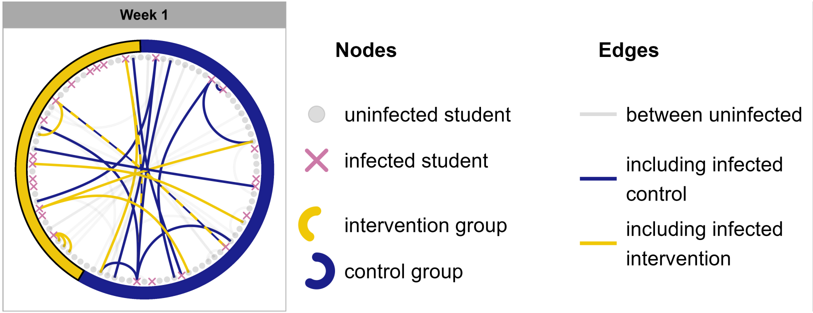
  is the **cumulative distribution function** (CDF) of T
- Reject $H_0^\sharp$ if $p^\sharp \leq 0.05$



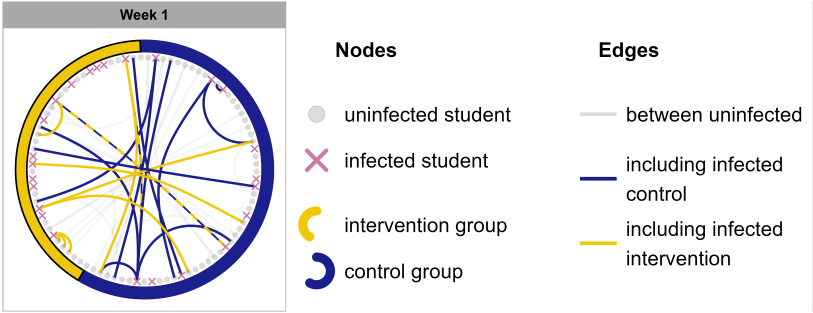
Choice of Test Statistic

- 93 (out of 579) students with at least one infection



Choice of Test Statistic

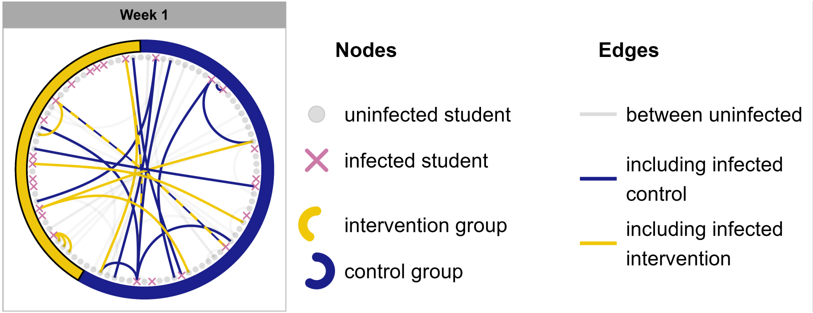
- 93 (out of 579) students with at least one infection
- Bold edges represent possible transmission events



Choice of Test Statistic

- 93 (out of 579) students with at least one infection
- Bold edges represent possible transmission events
  - Proportion of possible transmission events attributable to students in the intervention group:

$$T = \frac{\text{number of yellow edges}}{\text{total number of edges}} = \frac{\sum_{k=2}^{\tau} \sum_{i=1}^{n} \sum_{j \neq i} Z_i Y_i^{k-1} A_{ij}^{k-1}}{\sum_{k=2}^{\tau} \sum_{i=1}^{n} \sum_{j \neq i} Y_i^{k-1} A_{ij}^{k-1}} = 0.359$$



eX-FLU: Hypothesis Test Results

- Test statistic (proportion of possible transmission events attributable to students in the intervention group): $T = 0.359$.

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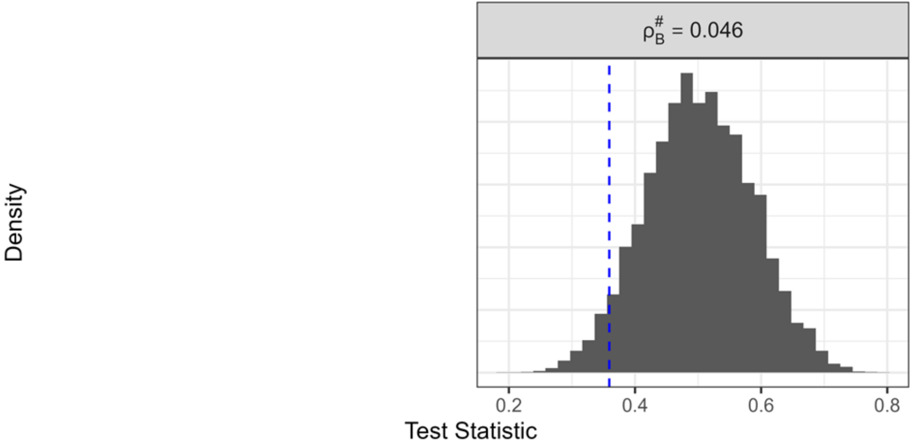

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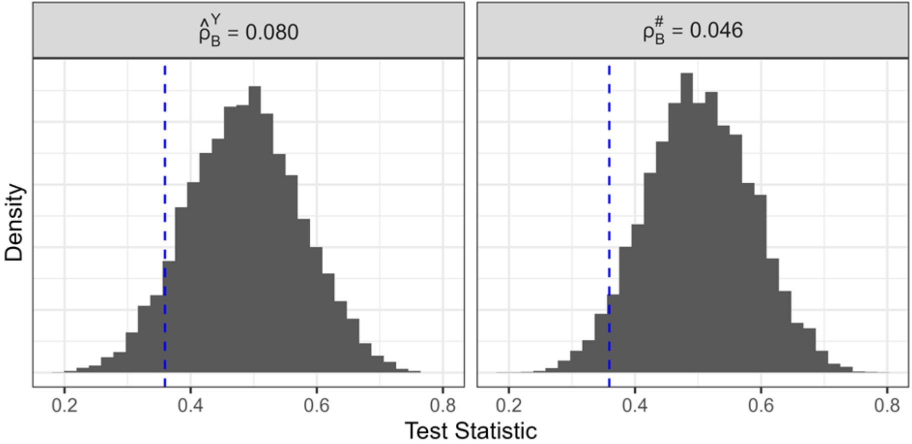

References I

Allison E. Aiello, Amanda M. Simanek, Marisa C. Eisenberg, Alison R. Walsh, Brian Davis, Erik Volz, Caroline Cheng, Jeanette J. Rainey, Amra Uzicanin, Hongjiang Gao, Nathaniel Osgood, Dylan Knowles, Kevin Stanley, Kara Tarter, and Arnold S. Monto. Design and methods of a social network isolation study for reducing respiratory infection transmission: The eX-FLU cluster randomized trial. *Epidemics*, 15:38–55, June 2016. ISSN 17554365. doi: 10.1016/j.epidem.2016.01.001. URL https://linkinghub.elsevier.com/retrieve/pii/S1755436516000025.

Shaina J. Alexandria, Michael G. Hudgens, and Allison E. Aiello. Assessing Intervention Effects in a Randomized Trial Within a Social Network. *Biometrics*, 79(2):1409–1419, June 2023. ISSN 0006-341X, 1541-0420. doi: 10.1111/biom.13606. URL https://academic.oup.com/biometrics/article/79/2/1409-1419/7513977.

R. A. Fisher. The design of experiments. *Nature*, 137(3459):252–254, February 1936. ISSN 0028-0836, 1476-4687. doi: 10.1038/137252a0.

Michael G Hudgens and M. Elizabeth Halloran. Toward causal inference with interference. *Journal of the American Statistical Association*, 103(482):832–842, 2008. ISSN 0162-1459, 1537-274X. doi: 10.1198/016214508000000292.

David M. Ritzwoller, Joseph P. Romano, and Azeem M. Shaikh. Randomization inference: Theory and applications, 2024.

Yao Zhang and Qingyuan Zhao. What is a randomization test? *Journal of the American Statistical Association*, 118(544):2928–2942, October 2023. ISSN 0162-1459, 1537-274X. doi: 10.1080/01621459.2023.2199814.

P.N. Zivich, M.C. Eisenberg, A.S. Monto, A. Uzicanin, R. S. Baric, T. P. Sheahan, J. J. Rainey, H. Gao, and A. E. Aiello. Transmission of viral pathogens in a social network of university students: the eX-FLU study. *Epidemiology and Infection*, 148:e267, 2020. ISSN 0950-2688, 1469-4409. doi: 10.1017/S0950268820001806. URL https://www.cambridge.org/core/product/identifier/S0950268820001806/type/journal_article.

Supplementary Slides

Testing $H_0^\sharp$

"How unlikely are the observed data under $H_0^\sharp$?"

Testing $H_0^\sharp$

"How unlikely are the observed data under $H_0^\sharp$?" **Randomization test**

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- Choose a test statistic $T = T(\mathbf{Z}, \overline{\mathbf{A}}, \overline{\mathbf{Y}})$
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Randomization	**Networks**	**Infections**	**Test Statistic**
$\mathbf{z}_1$			
$\mathbf{z}_2$			
$\cdots$			
$\mathbf{z}_{\lvert\mathcal{Z}\rvert}$			

Testing $H_0^\sharp$

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- Choose a test statistic $T = T(\mathbf{Z}, \overline{\mathbf{A}}, \overline{\mathbf{Y}})$
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Randomization	Networks	Infections	Test Statistic
$\mathbf{z}_1$	$\overline{\mathbf{A}}(\mathbf{z}_1) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_1) = ?$	$T(\mathbf{z}_1, ?, ?)$
$\mathbf{z}_2$	$\overline{\mathbf{A}}(\mathbf{z}_2) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_1) = ?$	$T(\mathbf{z}_1, ?, ?)$
...	...	...	...
$\mathbf{z}_{\lvert\mathcal{Z}\rvert}$	$\overline{\mathbf{A}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) = ?$	$T(\mathbf{z}_{\lvert\mathcal{Z}\rvert}, ?, ?)$

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$\mathbf{z}_2$	$\overline{\mathbf{A}}(\mathbf{z}_2) = \overline{\mathbf{A}}$	$\overline{\mathbf{Y}}(\mathbf{z}_2) = \overline{\mathbf{Y}}$	$T(\mathbf{z}_2, \overline{\mathbf{A}}, \overline{\mathbf{Y}})$
...	...	...	...
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...	...	...	...
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Randomization	Networks	Infections	Test Statistic
$\mathbf{z}_1$	$\overline{\mathbf{A}}(\mathbf{z}_1) = \overline{\mathbf{A}}$	$\overline{\mathbf{Y}}(\mathbf{z}_1) = \overline{\mathbf{Y}}$	$T(\mathbf{z}_1, \overline{\mathbf{A}}, \overline{\mathbf{Y}})$
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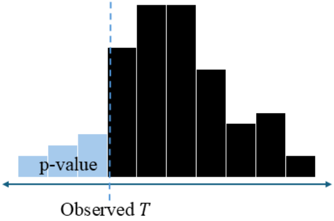

Testing $H_0^\sharp$

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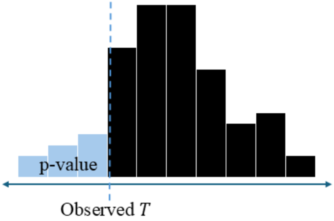
- Reject $H_0^\sharp$ if $\rho^\sharp \leq 0.05$



Testing $H_0^\sharp$

- This test of $H_0^\sharp$ controls the **type I error rate** exactly:

$$\Pr\left(\rho^\sharp \leq \alpha | H_0^\sharp\right) \leq \alpha$$

for any $\alpha \in [0, 1]$, for any sample size, and for any choice of test statistic

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  for any $\alpha \in [0, 1]$, for any sample size, and for any choice of test statistic

- The **power** of this test:

$$\Pr\left(\rho^\sharp \leq \alpha | H_1^\sharp\right)$$

  depends on the choice of test statistic

Testing H_0^Y

"How unlikely are the observed data under H_0^Y?"

Testing H_0^Y

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Randomization	Networks	Infections	Test Statistic
$\mathbf{z}_1$			
$\mathbf{z}_2$			
...			
$\mathbf{z}_{\lvert\mathcal{Z}\rvert}$			

Testing H_0^Y

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$\mathbf{z}_1$	$\overline{\mathbf{A}}(\mathbf{z}_1) =?$	$\overline{\mathbf{Y}}(\mathbf{z}_1) =?$	$T(\mathbf{z}_1, ?, ?)$
$\mathbf{z}_2$	$\overline{\mathbf{A}}(\mathbf{z}_2) =?$	$\overline{\mathbf{Y}}(\mathbf{z}_1) =?$	$T(\mathbf{z}_1, ?, ?)$
$\ldots$	$\ldots$	$\ldots$	$\ldots$
$\mathbf{z}_{\lvert\mathcal{Z}\rvert}$	$\overline{\mathbf{A}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) =?$	$\overline{\mathbf{Y}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) =?$	$T(\mathbf{z}_{\lvert\mathcal{Z}\rvert}, ?, ?)$

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    - This is not possible since $\overline{\mathbf{A}}(\mathbf{z})$ are **not imputable** under $H_0^Y \implies H_0^Y$ is **nonsharp**

Randomization	Networks	Infections	Test Statistic
$\mathbf{z}_1$	$\overline{\mathbf{A}}(\mathbf{z}_1) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_1) = \overline{\mathbf{Y}}$	$T(\mathbf{z}_1, ?, \overline{\mathbf{Y}})$
$\mathbf{z}_2$	$\overline{\mathbf{A}}(\mathbf{z}_2) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_2) = \overline{\mathbf{Y}}$	$T(\mathbf{z}_2, ?, \overline{\mathbf{Y}})$
...	...	...	...
$\mathbf{z}_{\lvert\mathcal{Z}\rvert}$	$\overline{\mathbf{A}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) = ?$	$\overline{\mathbf{Y}}(\mathbf{z}_{\lvert\mathcal{Z}\rvert}) = \overline{\mathbf{Y}}$	$T(\mathbf{z}_{\lvert\mathcal{Z}\rvert}, ?, \overline{\mathbf{Y}})$

Testing H_0^Y

- **Problem**: H_0^Y is nonsharp

Testing H_0^Y

- **Problem**: H_0^Y is nonsharp
  $\implies$ we don't know $F_T(t|H_0^Y)$ and thus cannot compute a p-value testing H_0^Y

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Testing H_0^Y

- **Problem**: H_0^Y is nonsharp
  $\implies$ we don't know $F_T(t|H_0^Y)$ and thus cannot compute a p-value testing H_0^Y
- **Solution**: model the distribution of potential network histories
  - Define $q(\overline{\mathbf{a}}, \mathbf{z}) = \Pr\{\overline{\mathbf{A}}(\mathbf{z}) = \overline{\mathbf{a}}\}$, the **probability mass function** (PMF) of $\overline{\mathbf{A}}(\mathbf{z})$

Testing H_0^Y

- **Problem**: H_0^Y is nonsharp

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- **Proposition**: This procedure will **asymptotically** control the type I error rate

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Goal: model a network ($\mathbf{A}$) given covariates ($\mathbf{X}$)

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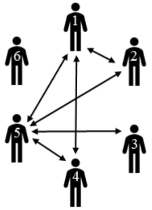

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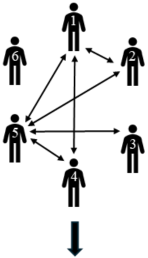

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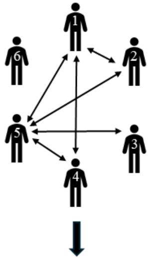

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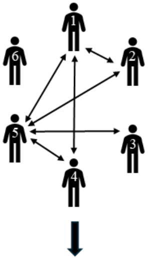

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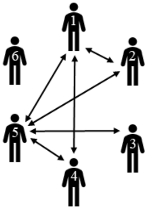

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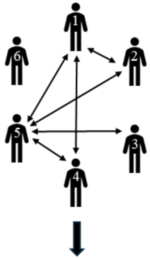

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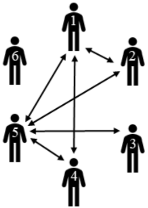

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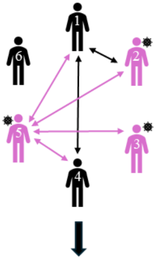

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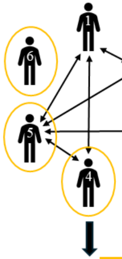

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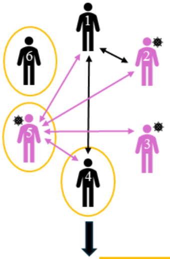
  - $g_{ZY}(\mathbf{a}, \mathbf{x}) = \sum_{i,j} a_{ij}(z_i y_i + z_j y_j)$



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Separable Temporal Exponential Family Random Graph Models

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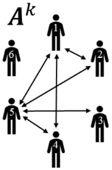

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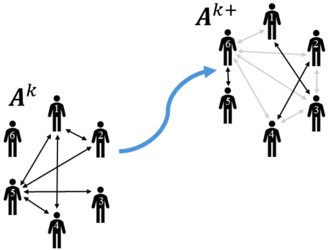
A **STERGM** assumes that, at each time step $k = 1, \ldots, \tau - 1$:

- New edges form according to a **formation** ERGM

$$\Pr_{\theta^+}(\mathbf{A}^{k+} = \mathbf{a}^{k+} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X}^{k+1} = \mathbf{x}^{k+1}) = \frac{\exp\left(\theta^+ \mathbf{g}^+(\mathbf{a}^{k+}, \mathbf{a}^k, \mathbf{x}^{k+1})\right)}{\kappa_{\mathbf{g}^+}\left(\theta^+, \mathcal{A}^+(\mathbf{a}^k), \mathbf{a}^k, \mathbf{x}^{k+1}\right)},$$



Separable Temporal Exponential Family Random Graph Models

A **STERGM** assumes that, at each time step $k = 1, \ldots, \tau - 1$:

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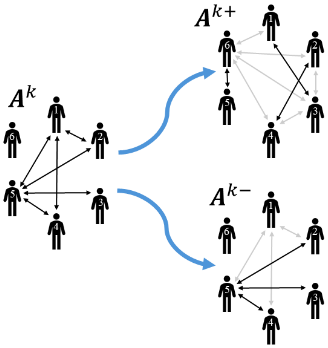
$$\Pr_{\theta^-}(\mathbf{A}^{k-} = \mathbf{a}^{k-} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X}^{k+1} = \mathbf{x}^{k+1}) = \frac{\exp\left(\theta^- \mathbf{g}^-(\mathbf{a}^{k-}, \mathbf{a}^k, \mathbf{x}^{k+1})\right)}{\kappa_{\mathbf{g}^-}(\theta^-, \mathcal{A}^-(\mathbf{a}^k), \mathbf{a}^k, \mathbf{x}^{k+1})},$$



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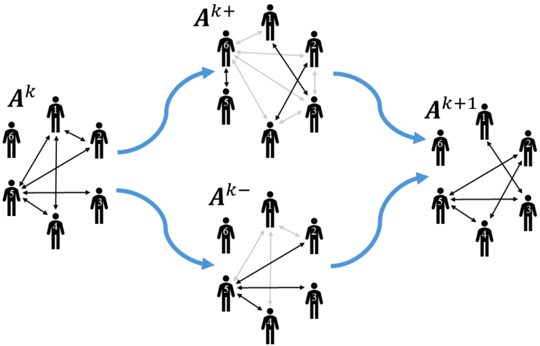
$$\mathbf{A}^{k+1} = \underbrace{\mathbf{A}^{k}}_{\text{previous network}} \cup \underbrace{(\mathbf{A}^{k+} - \mathbf{A}^{k})}_{\text{new edges formed}} - \underbrace{(\mathbf{A}^{k} - \mathbf{A}^{k-})}_{\text{old edges not persisting}}$$



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$$
\begin{tikzcd}
 & \mathbf{A}^{1+} \arrow[dr] & & \mathbf{A}^{2+} \arrow[dr] & & \mathbf{A}^{3+} \arrow[dr] & \\
\mathbf{A}^{1} \arrow[ur] \arrow[rr] \arrow[dr] & & \mathbf{A}^{2} \arrow[ur] \arrow[rr] \arrow[dr] & & \mathbf{A}^{3} \arrow[ur] \arrow[rr] \arrow[dr] & & \mathbf{A}^{4} \\
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2. the parameter space for $\theta = (\theta^+, \theta^-)$ is the product of the parameter spaces for θ^+ and θ^-

$$
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1. Estimate the unknown parameter θ in the PMF $q(\bar{\mathbf{a}}, \mathbf{z}; \theta)$ of $\overline{\mathbf{A}}(\mathbf{z})$

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   for $b = 1$ to B **do:**

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2. Approximate $F_T(t|H_0^Y)$ using B Monte-Carlo replicates:
   for $b = 1$ to B **do:**
   1. generate randomization vector $\mathbf{z}_b$ from $r(\mathbf{z})$
   2. generate networks $\bar{\mathbf{a}}_b$ from $q(\bar{\mathbf{a}}, \mathbf{z}_b; \widehat{\theta})$
   3. compute the test statistic $T_b = T(\mathbf{z}_b, \bar{\mathbf{a}}_b, \mathbf{Y})$

Testing H_0^Y

Testing Procedure:

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   2. generate networks $\bar{\mathbf{a}}_b$ from $q(\bar{\mathbf{a}}, \mathbf{z}_b; \widehat{\theta})$
   3. compute the test statistic $T_b = T(\mathbf{z}_b, \bar{\mathbf{a}}_b, \mathbf{Y})$
3. compute the plug-in p-value testing H_0^Y:

$$\widehat{\rho}_B^Y = B^{-1} \sum_{b=1}^{B} \mathbb{1}(T_b \leq T)$$

Type I Error Control

- **Proposition 2.1**: Let T_N be a test statistic with CDF F_N under H_0.
  1. Under H_0, $F_N(T_N)$ stochastically dominates a Uniform$(0, 1)$ distribution for any N.
  2. If the test statistic T_N has a continuous limiting distribution, then $F_N(T_N) \to^d$ Uniform$(0, 1)$.
- **Corollary 2.2**:
  1. Under $H_0^\sharp$, the sharp null p-value $\rho_N^\sharp$ stochastically dominates a Uniform$(0, 1)$ distribution for any N.
  2. Under H_0^Y, the oracle p-value ρ_N^Y stochastically dominates a Uniform$(0, 1)$ distribution for any N.
  3. If the test statistic T_N has a continuous limiting distribution, then $\rho_N^\sharp \to^d$ Uniform$(0, 1)$ under $H_0^\sharp$ and $\rho_N^Y \to^d$ Uniform$(0, 1)$ under H_0^Y.

Type I Error Control

- **Proposition 2.3**: Let $q(\mathbf{a}, \mathbf{z}, \theta) \equiv \Pr\{\mathbf{A}(\mathbf{z}) = \mathbf{a}; \theta\}$ denote the PMF of the distribution of stochastic potential networks $\mathbf{A}(\mathbf{z})$ at parameter value θ, let $\widehat{\theta}_N$ denote the estimator of θ, and let $F_N(\cdot; \theta)$ denote the CDF of the test statistic T_N at θ, with limiting CDF $F(\cdot; \theta)$. Let θ_0 denote the true value of θ. Assume the following:

  (A1) $\widehat{\theta}_N \to^p \theta_0$

  (A2) $F(t; \theta_0)$ is continuous in t on $\mathbb{R}$

  (A3) there exists a $\delta_0 > 0$ such that

  $$\sup_{\theta \in B_{\delta_0}(\theta_0)} \sup_{t \in \mathbb{R}} |F_N(t; \theta) - F(t; \theta)| \to 0$$

  (A4) $F(t; \theta)$ is continuous in θ at θ_0 uniformly in t, i.e.,

  $$\lim_{\theta \to \theta_0} \sup_{t \in \mathbb{R}} |F(t; \theta) - F(t; \theta_0)| = 0$$

  Then the plug-in p-value $\widehat{\rho}_N^Y$ converges in distribution to Uniform$(0, 1)$.

Type I Error Control

- **Proposition 2.4**: Let $\rho_N = F_N(T_N)$ for test statistic T_N and CDF F_N (not necessarily the true CDF of T_N). Let $T_N^* = h_N(T_N)$ for a sequence of deterministic, strictly increasing functions h_N. Define $F_N^*(t) = F_N\{h_N^{-1}(t)\}$, the (not necessarily true) CDF of the transformed test statistic, and let $\rho_N^* = F_N^*(T_N^*)$. Then $\rho_N^* = \rho_N$.
- **Corollary 2.5**: Let h_N be a sequence of deterministic, strictly increasing functions.
  1. If the hypotheses of Proposition 2.1 are met for a test statistic T_N, then the results also hold for $T_N^* = h_N(T_N)$.
  2. If the hypotheses of Proposition 2.3 are met for $T_N, F_N(\cdot;\theta)$, then the results also hold for $T_N^* = h_N(T_N), F_N^*(\cdot;\theta) = F_N\{h_N^{-1}(\cdot);\theta\}$.

Simulation Study

Goal: use simulated data to investigate the performance of the testing procedure

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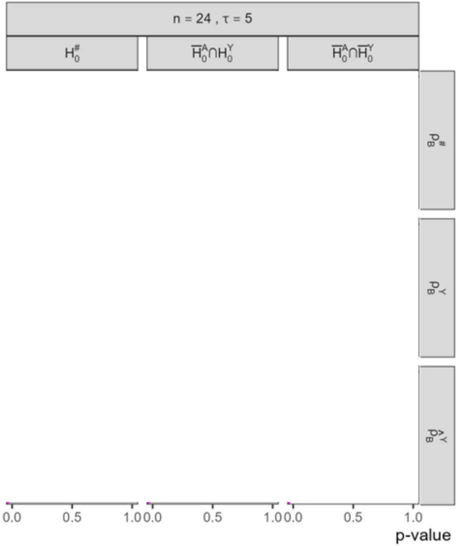
- Three testing procedures:
  1. $\rho_B^\sharp$: testing $H_0^\sharp$
  2. ρ_B^Y: testing H_0^Y using known q
  3. $\widehat{\rho}_B^Y$: testing H_0^Y using estimated q

Simulation Results: Empirical CDF of P-Values



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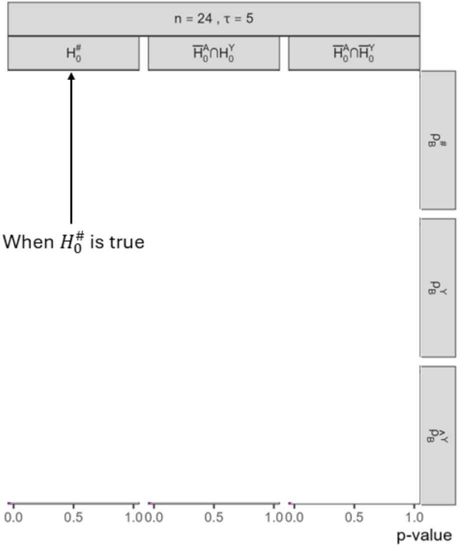

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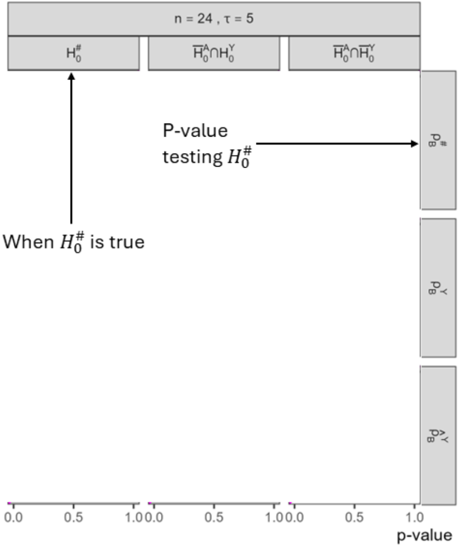

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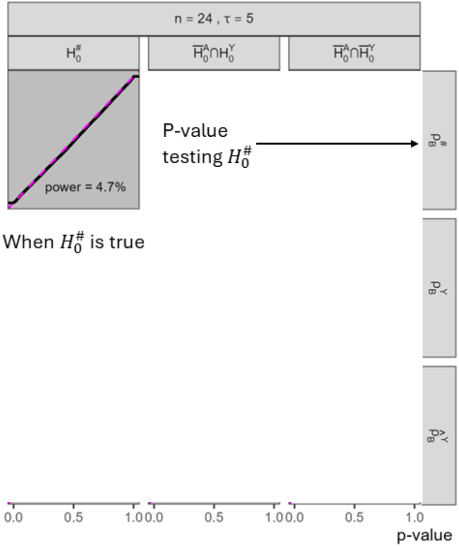

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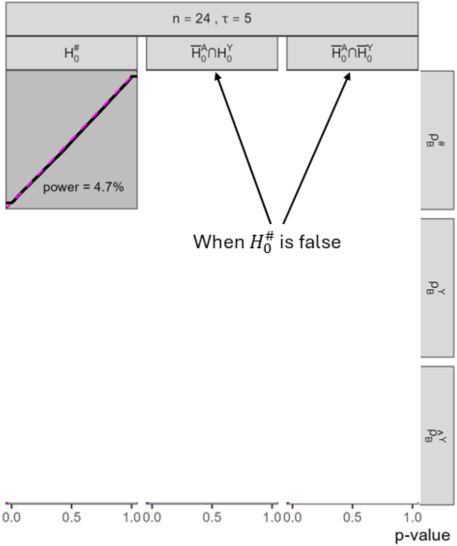

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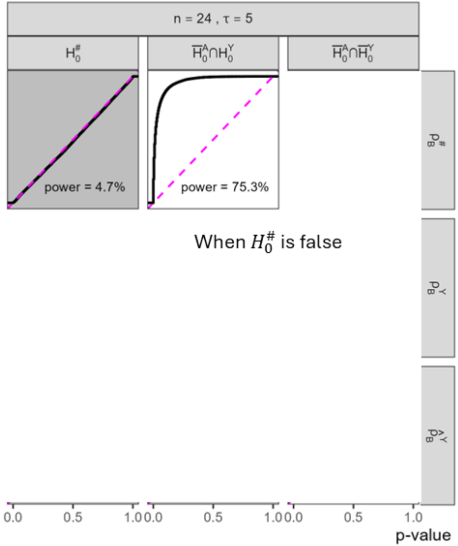

When $H_0^\#$ is false

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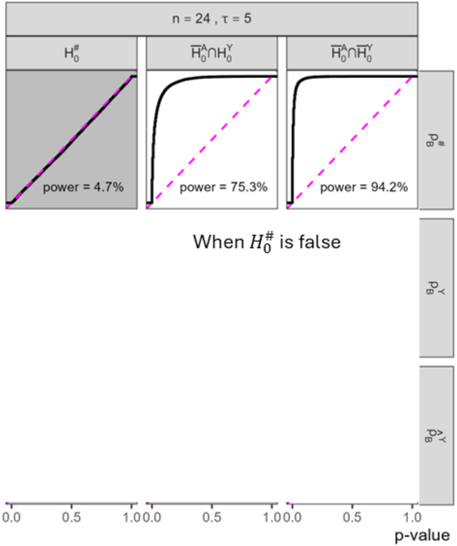

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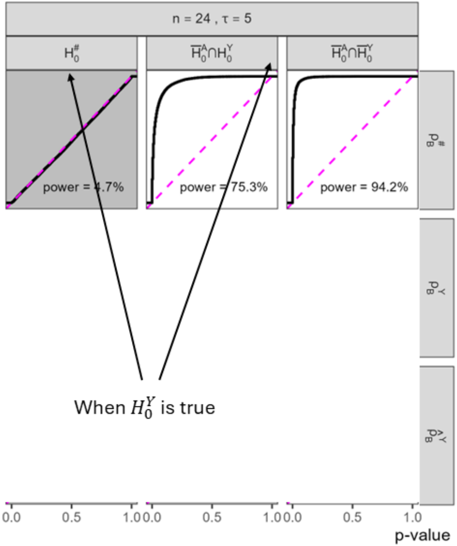

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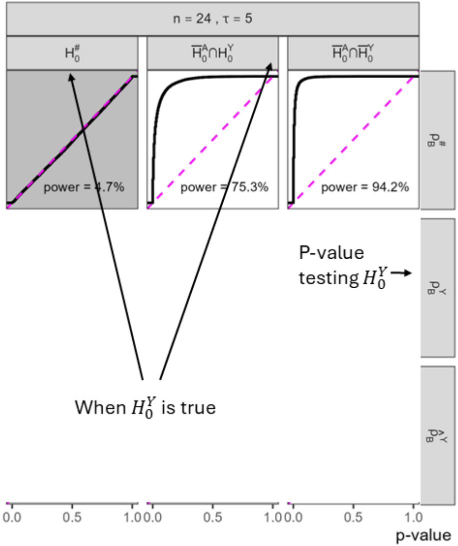

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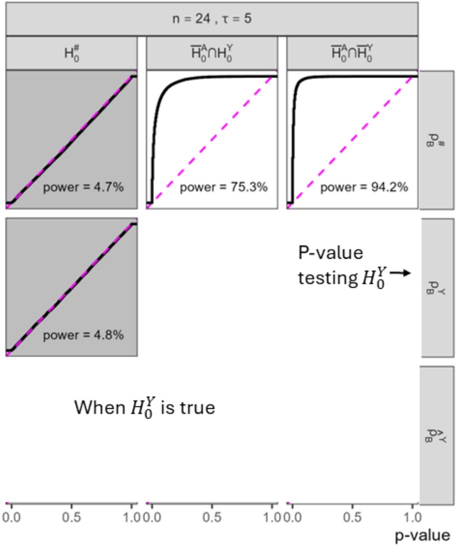

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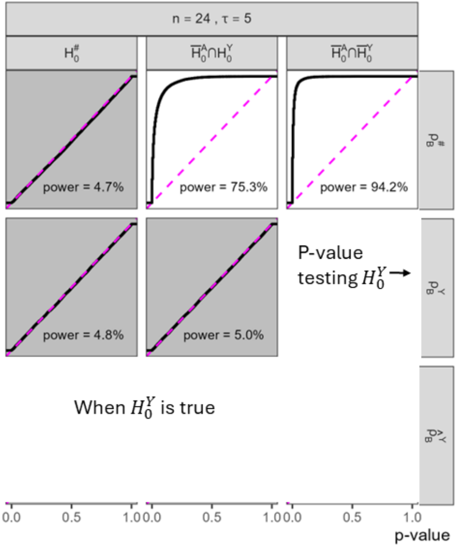

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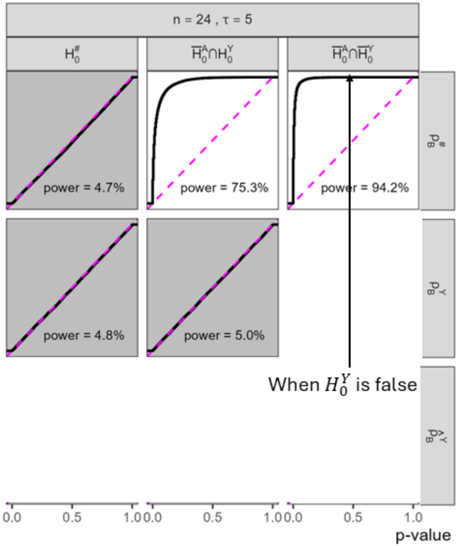

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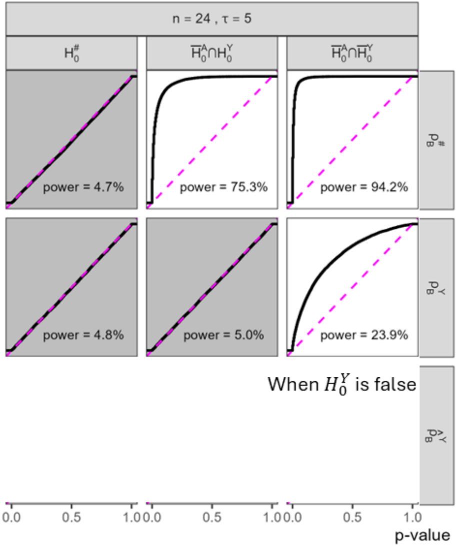

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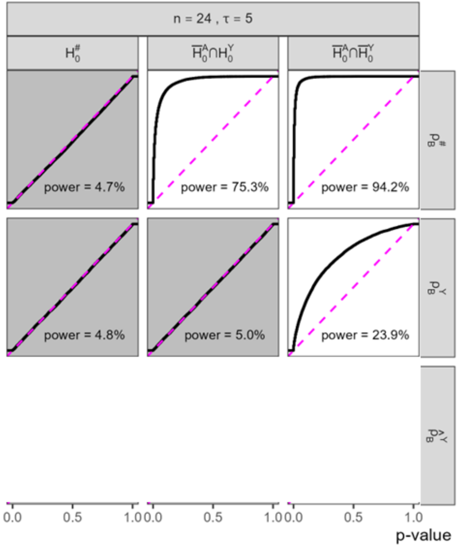

Using oracle knowledge of q

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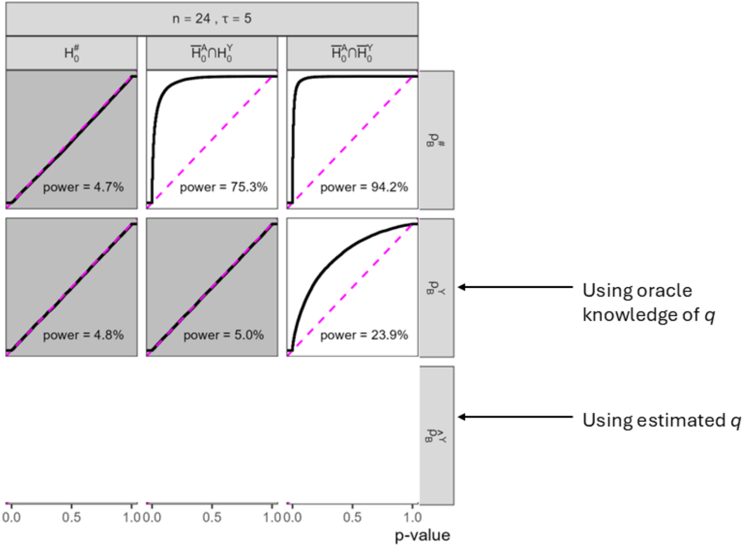

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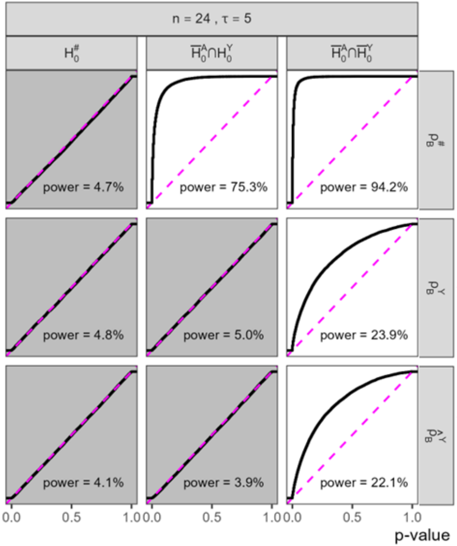
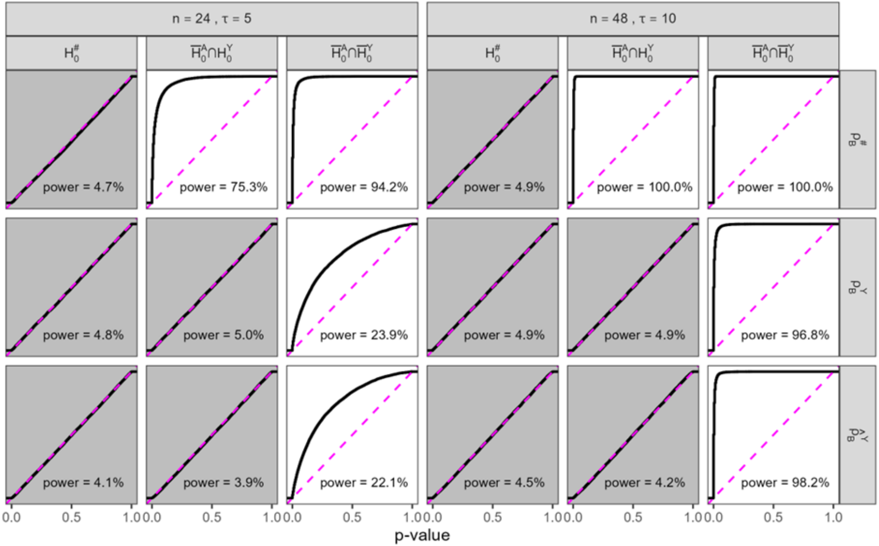


Using oracle knowledge of q

Using estimated q

Simulation Results: Empirical CDF of P-Values



eX-FLU: STERGM Specification

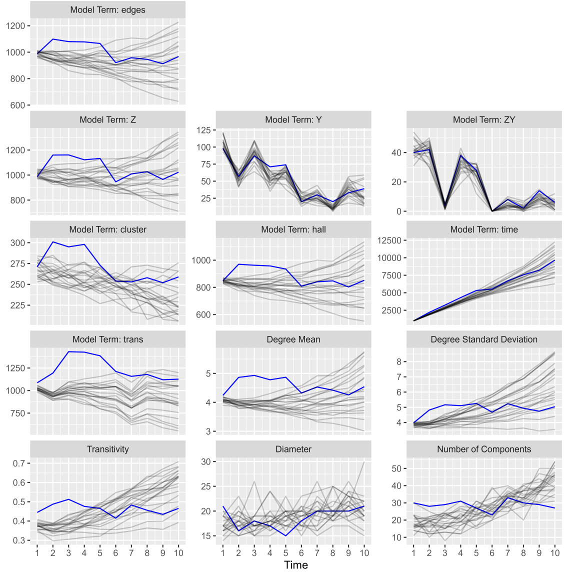
$$\mathbf{g}^{+}(\mathbf{a}^{k+}, \mathbf{a}^{k}, \mathbf{x}^{k+1}) =$$

$$\begin{bmatrix} g^{+}_{\text{edges}}(\mathbf{a}^{k+}, \mathbf{x}^{k+1}) \\ g^{+}_{Z}(\mathbf{a}^{k+}, \mathbf{x}^{k+1}) \\ g^{+}_{Y}(\mathbf{a}^{k+}, \mathbf{x}^{k+1}) \\ g^{+}_{ZY}(\mathbf{a}^{k+}, \mathbf{x}^{k+1}) \\ g^{+}_{\text{room}}(\mathbf{a}^{k+}, \mathbf{x}^{k+1}) \\ g^{+}_{\text{cluster}}(\mathbf{a}^{k+}, \mathbf{x}^{k+1}) \\ g^{+}_{\text{hall}}(\mathbf{a}^{k+}, \mathbf{x}^{k+1}) \\ g^{+}_{\text{time}}(\mathbf{a}^{k+}, \mathbf{x}^{k+1}) \\ g^{+}_{\text{trans}}(\mathbf{a}^{k}, \mathbf{x}^{k+1}) \end{bmatrix} = \begin{bmatrix} \sum_{i,j} a_{ij}^{k+} \\ \sum_{i,j} a_{ij}^{k+}(z_i + z_j) \\ \sum_{i,j} a_{ij}^{k+}(y_i^{k+1} + y_j^{k+1}) \\ \sum_{i,j} a_{ij}^{k+}(z_i y_i^{k+1} + z_j y_j^{k+1}) \sum_{i,j} a_{ij}^{k+}\mathbb{1}(i,j \text{ roommates}) \\ \sum_{i,j} a_{ij}^{k+}\mathbb{1}(i,j \text{ cluster-mates}) \\ \sum_{i,j} a_{ij}^{k+}\mathbb{1}(i,j \text{ hall-mates}) \\ \sum_{i,j} a_{ij}^{k+} \\ \sum_{i,j} \log\left(1 + \sum_{l} a_{il}^{k} a_{jl}^{k}\right) \end{bmatrix}$$

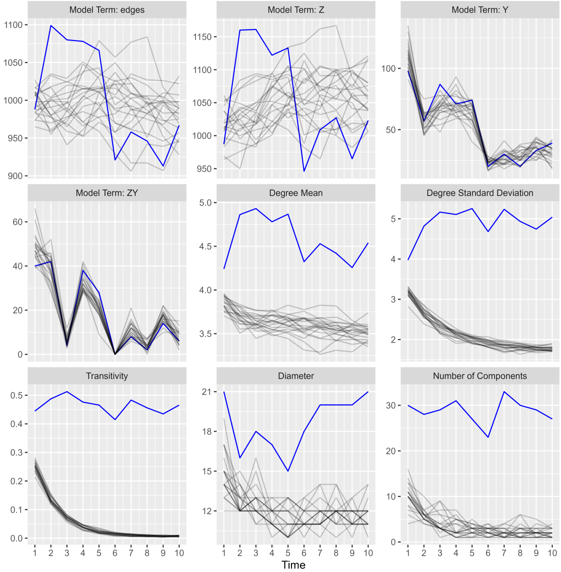
eX-FLU: STERGM Results

Model	Parameter	Estimate (95% Confidence Interval)
Formation	$\theta^{+}_{\text{edges}}$	-7.85 (-8.00, -7.71)
	θ^{+}_{Z}	0.09 (0.02, 0.15)
	θ^{+}_{Y}	0.71 (0.50, 0.93)
	θ^{+}_{ZY}	-0.42 (-0.79, -0.05)
	θ^{+}_{room}	∞ (offset)
	$\theta^{+}_{\text{cluster}}$	2.02 (1.90, 2.14)
	θ^{+}_{hall}	1.71 (1.57, 1.84)
	θ^{+}_{time}	-0.09 (-0.11, -0.07)
	$\theta^{+}_{\text{trans}}$	2.88 (2.82, 2.94)
Persistence	$\theta^{-}_{\text{edges}}$	0.16 (-0.02, 0.33)
	θ^{-}_{Z}	0.03 (-0.03, 0.10)
	θ^{-}_{Y}	-0.06 (-0.32, 0.21)
	θ^{-}_{ZY}	-0.11 (-0.55, 0.33)
	θ^{-}_{room}	∞ (offset)
	$\theta^{-}_{\text{cluster}}$	0.26 (0.13, 0.39)
	θ^{-}_{hall}	0.15 (0.00, 0.31)
	θ^{-}_{time}	0.04 (0.02, 0.06)
	$\theta^{-}_{\text{trans}}$	0.42 (0.35, 0.48)

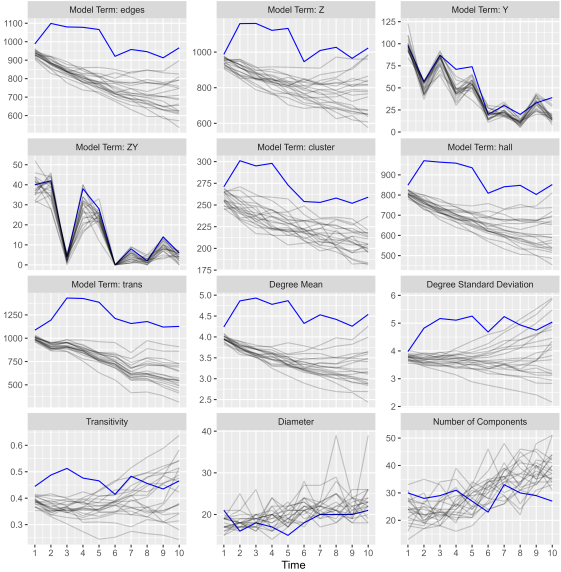
eX-FLU Diagnostics: STERGM



eX-FLU Diagnostics: Alternate STERGM 1



eX-FLU Diagnostics: Alternate STERGM 2



eX-FLU Sensitivity Analysis